

ADVANCED DATABASE MANAGEMENT PROJECT REPORT

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ABSTRACT

Effective data modeling plays a pivotal role in aligning business processes with technical constraints. This project undertakes the creation and optimization of a relational data model for an Order Management system, emphasizing the development of tables, implementation of integrity constraints, and efficient data loading. Subsequently, a series of SQL queries are formulated, exploring optimization techniques and the application of triggers to enhance data accuracy. The project concludes with the creation of insightful dashboards using Power BI, offering a comprehensive visual representation of key metrics.

1. INTRODUCTION

The efficient management and analysis of organizational data play an important role in informed decision-making and strategic planning. In this context, our report encompasses a comprehensive exploration of a relational data model, focusing on tables related to purchase orders, vendors, and products. The dataset provided includes PurchaseOrderHeader.csv, PurchaseOrderDetail.csv, Vendor.csv, and ProductVendor.csv, each contributing to the intricate web of relationships within the system.

Our primary objectives revolve around the creation of tables, implementation of integrity constraints, and the development of a data-loading tool to populate these tables. Additionally, we delve into SQL queries, aiming not only to extract valuable insights but also to optimize query performance through the use of indexes and materialized views.

Furthermore, we extend our analysis with triggers, proposing two key triggers aimed at maintaining transaction history and ensuring data consistency within the PurchaseOrderHeader and PurchaseOrderDetail tables.

The latter part of our exploration exploits Power BI to generate dashboards, facilitating the presentation of critical business metrics. These dashboards serve as dynamic tools for monitoring and analyzing data trends, offering insights into top-selling products and vendor performance metrics.

2. DATA DESCRIPTION

PurchaseOrderDetail.csv:

- purchaseorderid: Identifier for the purchase order.
- purchaseorderdetailid: Unique identifier for the purchase order detail.
- duedate: Date when the order is due.
- orderqty: Quantity of the ordered product.
- productid: Identifier for the product.
- unitprice: Unit price of the product.
- receivedqty: Quantity of products received.
- rejectedqty: Quantity of products rejected.
- modifieddate: Date of the last modification.

PurchaseOrderHeader.csv:

- purchaseorderid: Identifier for the purchase order.
- revisionnumber: Revision number of the purchase order.
- status: Status of the purchase order.
- employeeid: Identifier for the employee associated with the order.
- vendorid: Identifier for the vendor associated with the order.
- shipmethodid: Identifier for the shipping method.
- orderdate: Date when the order was placed.
- shipdate: Date when the order is scheduled to be shipped.
- subtotal: Subtotal amount of the purchase order.
- taxamt: Tax amount associated with the order.
- freight: Freight cost associated with the order.
- modifieddate: Date of the last modification.

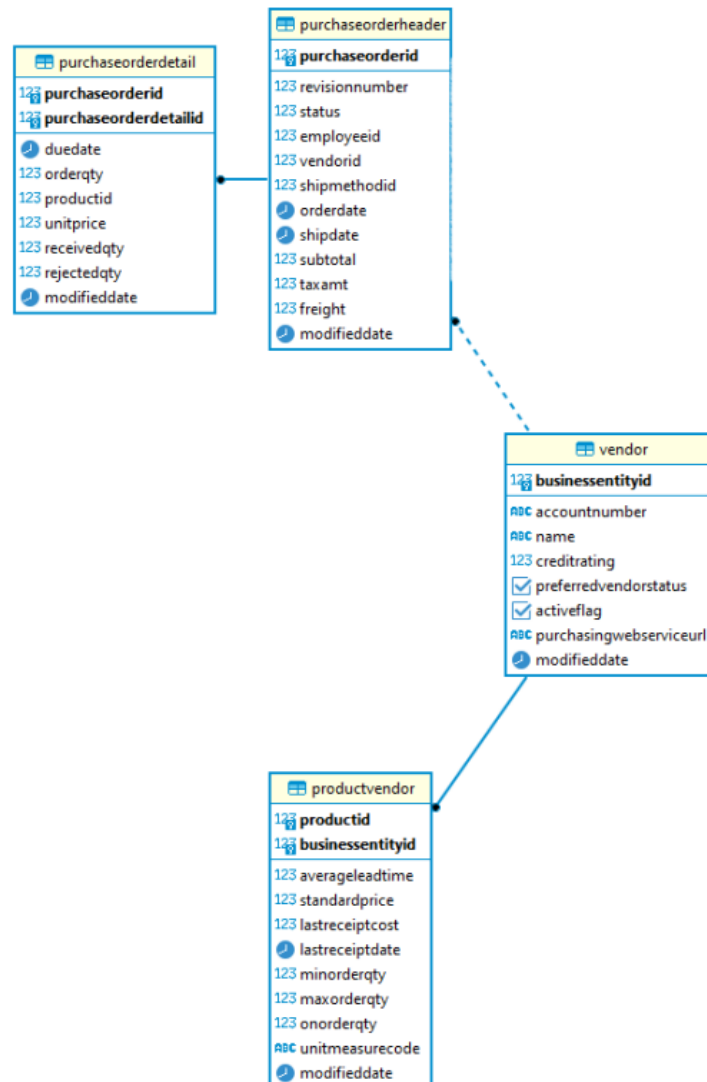
Vendor.csv:

- businessentityid: Identifier for the vendor's business entity.
- accountnumber: Account number associated with the vendor.
- name: Name of the vendor.
- creditrating: Credit rating of the vendor.
- preferredvendorstatus: Status indicating vendor preference.
- activeflag: Flag indicating the vendor's active status.
- purchasingwebserviceurl: URL for the vendor's purchasing web service.
- modifieddate: Date of the last modification.

ProductVendor.csv:

- productid: Identifier for the product.
- businessentityid: Identifier for the vendor's business entity.
- averageleadtime: Average lead time for the product from the vendor.
- standardprice: Standard price of the product.
- lastreceiptcost: Cost of the last received product.
- lastreceiptdate: Date of the last product receipt.
- minorderqty: Minimum order quantity for the product.
- maxorderqty: Maximum order quantity for the product.
- onorderqty: Quantity of the product on order.
- unitmeasurecode: Code indicating the unit of measure for the product.
- modifieddate: Date of the last modification.

The Order Management relational model:



3. DATA CLEANING

Prior to loading data, it's essential to ensure that the data is properly formatted and aligned with the defined table structures.

In order to ensure the accuracy and consistency of the CSV data, we implemented a robust cleaning process that addressed two key aspects: removing spaces surrounding data and handling missing values in numeric columns.

Here some examples of these problems:

579	246	579	2012-10-05 00:00:00	3	423	45.423	3.00	0.00	2012-09-28 00:00:00.000
580	246	579	2012-10-05 00:00:00	3	423	45.423	3.00	0.00	2012-09-28 00:00:00.000
581									
582	246	580	2012-10-05 00:00:00	3	424	49.6965	3.00	0.00	2012-09-28 00:00:00.000
583	246	581	2012-10-05 00:00:00	3	425	45.423	3.00	0.00	2012-09-28 00:00:00.000
584	247	582	2012-10-05 00:00:00	3	320	47.4705	3.00	0.00	2012-09-28 00:00:00.000
585	247	583	2012-10-05 00:00:00	3	321	47.708	3.00	0.00	2012-09-28 00:00:00.000

```
productid          4
businessentityid   1650
averageleadtime    17
standardprice      54.31
lastreceiptcost     57.0255
lastreceiptdate     2011-08-29 00:00:00.000
minorderqty        1
maxorderqty        5
onorderqty
unitmeasurecode    CTN
modifieddate        2011-08-29 00:00:00.000
Name: 2, dtype: object
```

In fact, we noticed there were spaces around some of the values, potentially causing inconsistencies and issues in downstream analyses.

To address this, we applied regex-based string manipulation using the `str.strip()` method in Pandas. This method removes leading and trailing whitespaces from string values, ensuring uniformity across the dataset. The operation looked something like this:

```
[ ] colonne_index = 9
    df.iloc[:, colonne_index] = df.iloc[:, colonne_index].str.strip()
```

We also tackled the issue of missing values in numeric columns. It's crucial to handle these gaps in the data to avoid disruptions in calculations and visualizations. Leveraging regex, we replaced empty values with 0 in all numeric columns. Here's an example:

```
colonne_index = 8
df.iloc[:, colonne_index] = df.iloc[:, colonne_index].replace(r'^\s*$',0, regex=True)
```

By combining regex and Pandas in Google Colab, we successfully cleaned our CSV data, removing extraneous spaces and addressing missing values in numeric columns. This enhanced dataset is now better prepared for analysis and visualization, contributing to more accurate and reliable insights.

4. TABLE CREATION AND DATA LOADING

1. Table Creation

In the creation of the database tables, specific data types were chosen for each column to optimize storage and ensure accurate representation of the data. For the "purchaseorderdetail" table, integers were selected for identifiers, timestamps for date-related information, and numeric/decimal types for quantities and prices. Constraints such as "not null" were applied to essential columns like "unitprice," "receivedqty," and "rejectedqty" to enforce mandatory data entry.

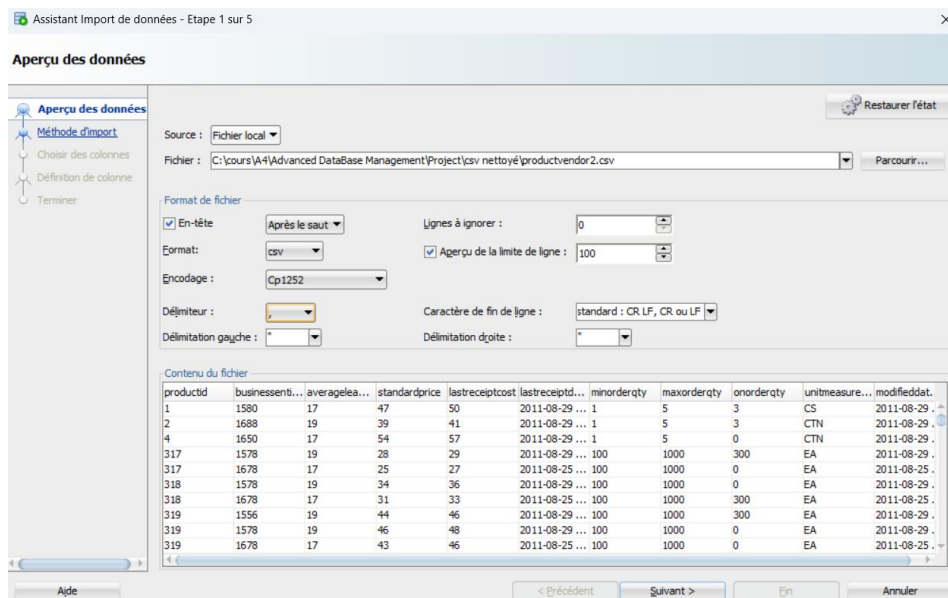
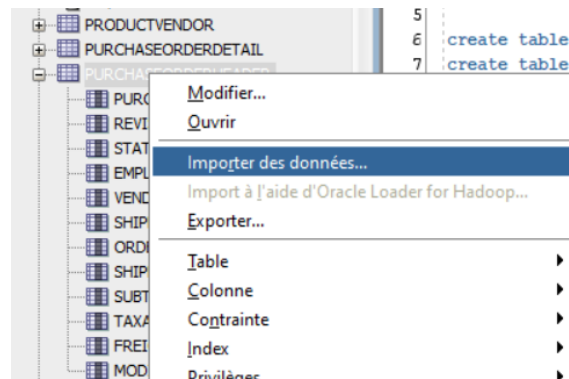
Similarly, the "purchaseorderheader" table utilizes integers for identifiers, timestamps for dates, and numeric types for financial values. The "vendor" table employs integers for identifiers, varchar2 for textual data, and a timestamp to track modifications. The "productvendor" table uses integers for identifiers, timestamps for dates, and a mix of integers, numerics, and varchar2 for different types of information.

Foreign key constraints were established to link tables. For instance, the "fk_purchaseorderdetailid" links "purchaseorderdetail" to "purchaseorderheader," ensuring data integrity between these related tables. Additionally, "fk_businessentityid" establishes a link between "productvendor" and "vendor."

In summary, the chosen data types and constraints were tailored to the nature of the information each column represents, aiming to maintain data accuracy, consistency, and overall database integrity.

2. Data loading

SQL Developer provides a user-friendly Import Wizard that simplifies the process of loading data into tables. This tool allows you to choose the source of your data and map it to the corresponding database tables. You can see in the figures below how we proceeded.



This method creates an SQL Script that allowed us to import all the data contained in the csv files. It's also worth noting that because of the unique constraints we defined, we need to import some data into the tables before the others to avoid unique constraint violations.

5. SQL QUERIES

The SQL queries and triggers presented showcase a thoughtful approach to optimizing data retrieval and ensuring data integrity within the database. In the optimized queries, indexes were

strategically created on relevant columns to enhance the performance of specific operations. For instance, indexes were implemented on columns like `creditrating`, `productid`, `orderqty`, and `purchaseorderid`, contributing to more efficient data filtering and retrieval. These optimizations play a pivotal role in improving the overall responsiveness of the database when executing complex queries.

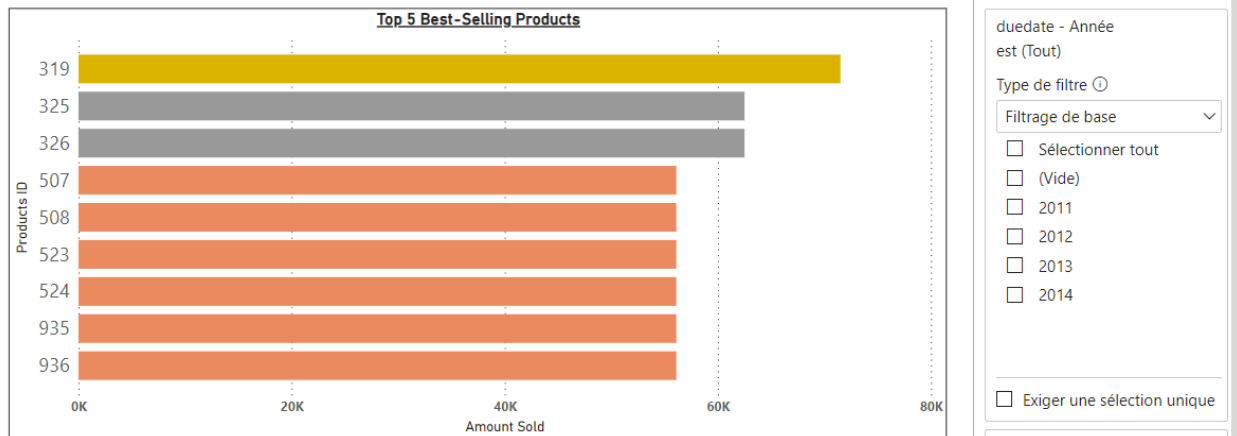
Additionally, the introduction of triggers adds an extra layer of control and consistency to the database operations. The trigger `trg_before_update_purchaseorderdetail` is triggered before an update on the `purchaseorderdetail` table. This trigger not only inserts a historical record into the `transaction_history` table but also updates the `ModifiedDate` and recalculates the `SubTotal` in the corresponding `PurchaseOrderHeader`. This ensures that any changes to purchase order details are not only logged but also reflected in the aggregated values at the header level.

The queries themselves are designed to extract information from the database. They cover a range of scenarios, from retrieving vendor-product relationships based on specific criteria to aggregating statistics about vendors and products. Aggregations, filtering conditions, and join operations are used to provide a comprehensive view of the relevant data.

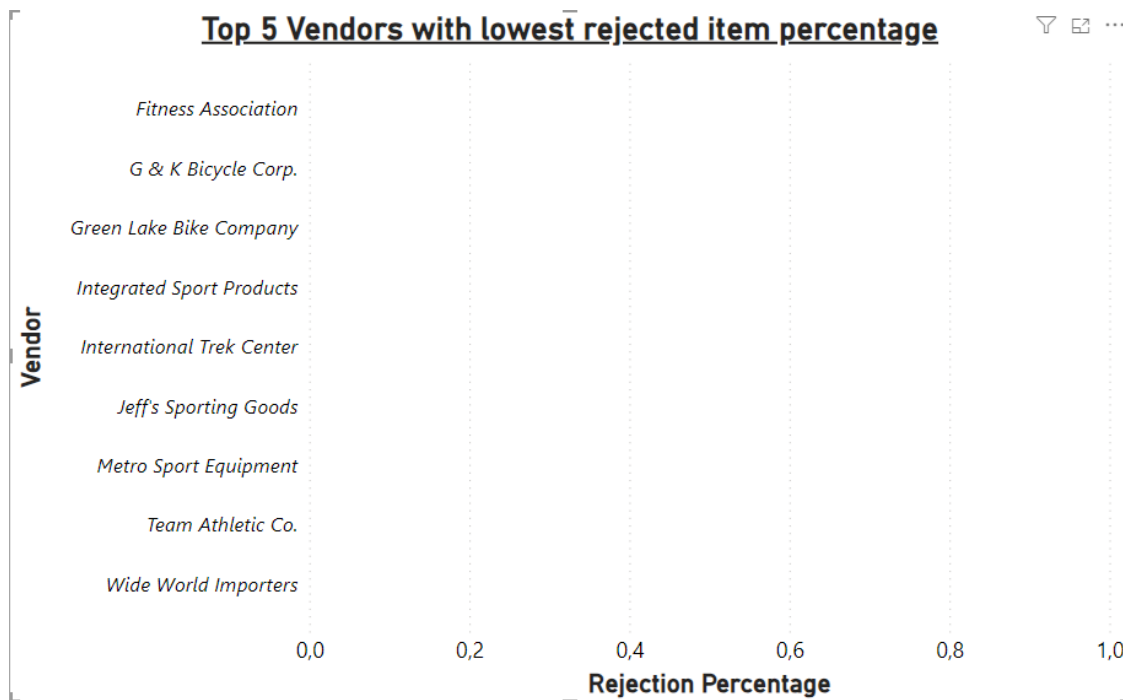
The goal of these optimizations and triggers is to create a well-performing and consistent database system. Indexing enhances query speed, while triggers enforce data consistency and historical tracking during updates. Together, these measures align with best practices for effective database management, ensuring that the database not only performs well but also maintains the accuracy and integrity of its data over time.

6. DATA VISUALIZATION

Power BI enables the creation of a dynamic dashboard so we used it in order to generate a dashboard that represents the Top 5 best-selling products in terms of quantity per year as well as one that represents The Top 5 vendors who have the lowest percentage of rejected received items and also the evolution of the amount of product 319 ordered from May 2011 to August 2014.

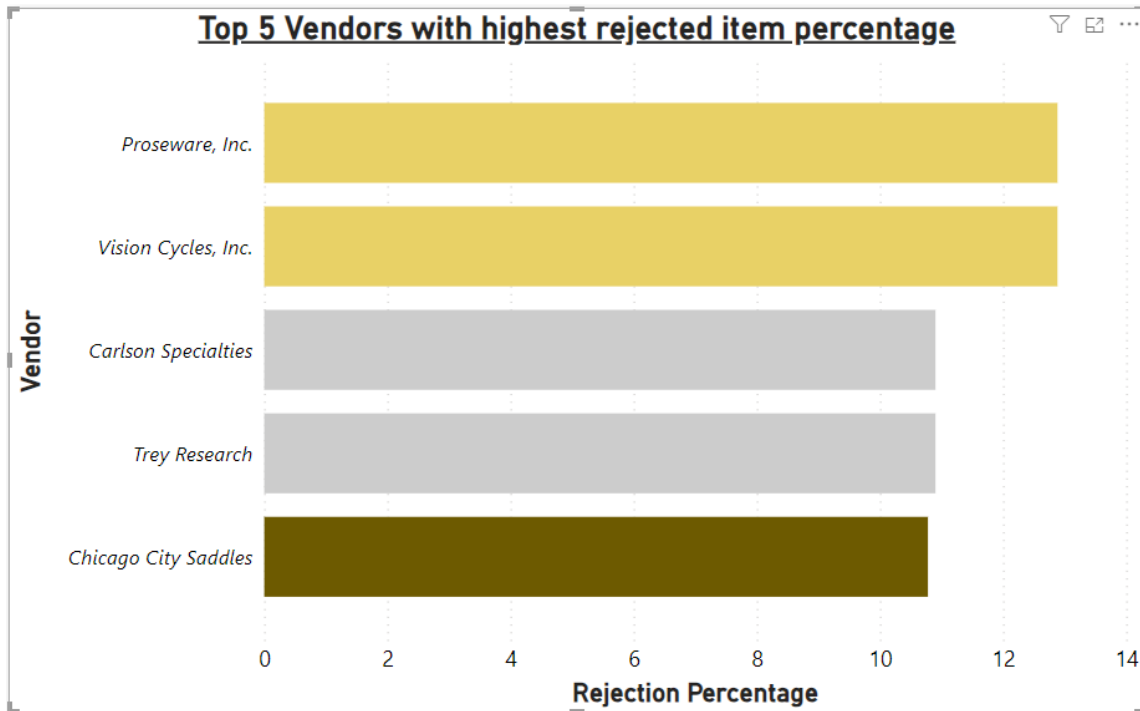


Here is a visualization representing the top 5 vendors with lowest rejected item percentage:

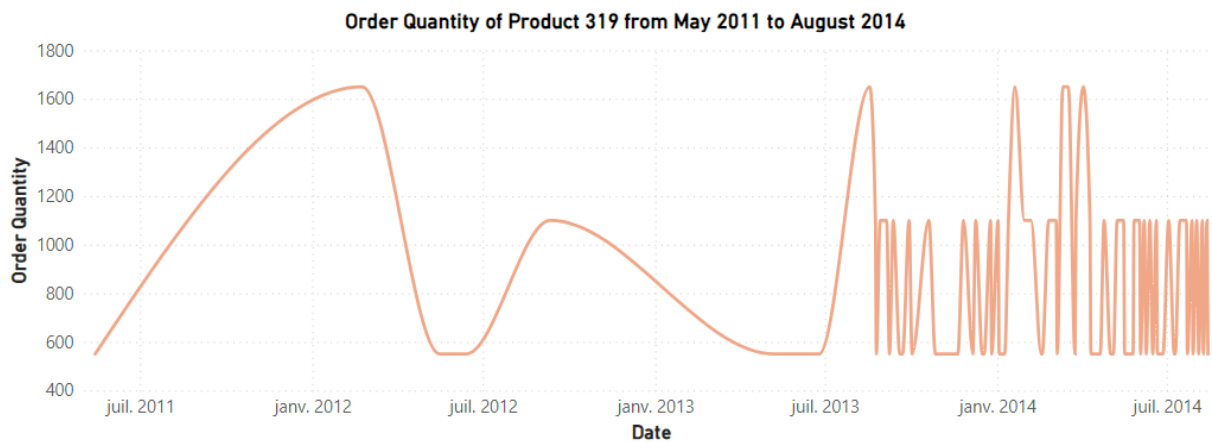


As we can see the top 5 vendors with the lowest received item rejection rate all have 0% rejection which is not interesting as a visualization. Thus we decided to go for another visualization.

Here is a visualization representing the top 5 vendors with highest rejected item percentage:



This visualization has been made on a dashboard composed of employed csv files during the project. We created this grouped thanks to the relations between the tables.



7. CONCLUSION

In summary, this project designed and optimized a relational data model for an Order Management system, ensuring data accuracy, consistency, and efficient retrieval. The implementation of SQL queries, triggers, and Power BI dashboards demonstrates a comprehensive approach to database management. The project's success lies in its ability to

translate theoretical concepts into practical solutions, providing a reliable foundation for informed decision-making and strategic insights.